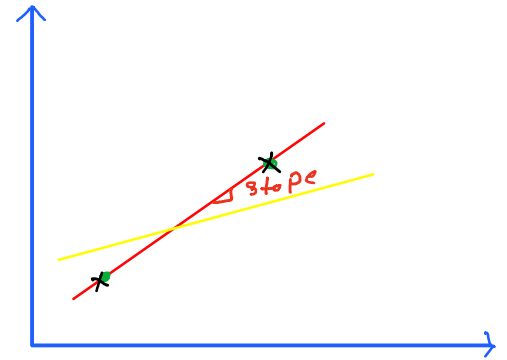


L2 (Ridge) Regularization

Monday, 22 July 2024 2:57 PM

$$\text{Cost Function} \rightarrow \frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

$$\begin{aligned} \text{Error} &\Rightarrow \text{Actual} - \text{Pred} \\ &= (\hat{y}_i - y_i)^2 \\ &= 0 \end{aligned}$$



in Ridge cost function

$$\begin{aligned} &= (\hat{y}_i - y_i)^2 + \lambda (\text{slope})^2 \\ &= 0 + 1(2)^2 \\ &= 4 \end{aligned}$$

take $\lambda = 1$

$$\Rightarrow \{\text{some Error}\} + 1(1.5)^2$$

take $\lambda = 1$

$$\approx 3.0$$